

♥ RESEARCH INTERESTS

- **Multimodal (Video) Generation / Understanding**: self-correcting and adaptive architecture with RL/post-training
- **Physical AI**: adaptive action modeling in embodied scenario with perceptual modality (e.g., 3D LiDAR, Gaze, HD Maps).

🏠 EDUCATION

The University of North Carolina at Chapel Hill

Ph.D. Computer Science (Advisor: Mohit Bansal)

NC, United States

Aug. 2024 —

Korea University

B.E. Department of Statistics

Seoul, South Korea

Mar. 2019 — Feb. 2024

🏢 RESEARCH EXPERIENCES

Meta Superintelligence Labs

Research Intern (Mentor: Markus Woodson, Tobias Hinz)

CA, United States

May.2026 —

Adobe Research

Research Intern (Mentor: Subhojyoti Mukherjee, Branislav Kveton)

CA, United States

May.2025 — Nov.2025

UNC Chapel Hill

Research Assistant (Advisor: Mohit Bansal)

NC, United States

Aug.2024 — Current

KAIST

Research Intern / Contract Researcher (Advisor: Sung Ju Hwang)

Seoul, South Korea

Mar.2023 — Aug.2024

NAVER LABS

Research Intern (Mentor: Minheok Heo)

Jungja, South Korea

Jul.2022 — Dec.2022

Korea University

Research Intern (Advisor: Jinkyu Kim)

Seoul, South Korea

Jul.2021 — Dec.2022

📖 PUBLICATIONS

[P3] VisionCoach: Reinforcing Grounded Video Reasoning vis Visual-Perception Prompting

[Daeun Lee](#), Shoubin Yu, Yue Zhang, Mohit Bansal.

Under review

[P2] IndustryNav: Exploring Spatial Reasoning of Embodied Agents in Dynamic Industrial Navigation

Yifan Li, Lichi Li, Anh Dao, Xinyu Zhou, Yicheng Qiao, Zheda Mai, [Daeun Lee](#), Zichen Chen, Zhen Tan, Mohit Bansal, Yu Kong.

Under review

[C7] VideoRepair: Self-Correcting Text-to-Video Generation with Misalignment Detection and Localized Refinement

[Daeun Lee](#), Jaehong Yoon, Jaemin Cho, Mohit Bansal.

The 64th Annual Meeting of the Association for Computational Linguistics (ACL) Findings, 2026.

[C6] StreamGaze: Gaze-Guided Temporal Reasoning and Proactive Understanding in Streaming Videos

[Daeun Lee](#), Subhojyoti Mukherjee, Brano Kveton, Ryan Rossi, Viet Lai, David Seunghyun Yoon, Franck Dernoncourt, Trung Bui, Mohit Bansal.

The IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.

[C5] Video-Skill-CoT: Skill-based Chain-of-Thoughts for Domain-Adaptive Video Reasoning

[Daeun Lee*](#), Jaehong Yoon*, Jaemin Cho, Mohit Bansal.

Empirical Methods in Natural Language Processing (EMNLP) Findings, 2025.

[C4] BECoTTA: Input-dependent Online Blending of Experts for Continual Test-Time Adaptation

[Daeun Lee*](#), Jaehong Yoon*, Sung Ju Hwang.

International Conference on Machine Learning (ICML), 2024

[C3] HD Maps are Lane Detection Generalizers: A Novel Generative Framework for Single-Source Domain Generalization

[Daeun Lee](#), Minhyeok Heo, Jiwon Kim.

CVPR Data-Driven Autonomous Driving Simulation Workshop (CVPRW), 2024

[C2] Resolving Class Imbalance for LiDAR-based Object Detector by Dynamic Weight Average and Contextual Ground Truth Sampling

[Daeun Lee](#), Jinkyu Kim.

IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2023

[C1] Bridging the Domain Gap towards Generalization in Automatic Colorization

Hyejin Lee, Daehee Kim, [Daeun Lee](#), Jinkyu Kim and Jaekoo Lee.

European Conference on Computer Vision (ECCV), 2022

[P1] Trajectory Prediction by Clustering Human Interactions at Multiple Scales

Chiho Choi*, [Daeun Lee*](#), Srikanth Malla, Sangjae Bae, Jinkyu Kim.

Preprint

 ACADEMIC SERVICES

Reviewer

CVPR 2022-2026, ECCV/ICCV 2022-2026, AAAI 2025, NeurIPS 2025

Invited Talks

Sep 2025. **SKKU (Sungkyunkwan University)** Google Developer Groups .

Jan 2025. **Cisco**. 'Reliable Text-to-Video Generation'

 AWARDS/HONORS

Travel grant from ICML 2024 Area Chair

Jun.2024

Digital Innovation Big Data Contest

May.2021

2nd place, Korea Enterprise Data Corp.(KED)

ICT Autonomous Driving Project

Dec.2020

5st place, The Federation of Korean Information Industries

Financial Big Data Festival

Dec.2020

1st place, MiraeAsset.Corp

Kakao Arena Competition

May.2020

Top 2%, Kakao.Corp

Outstanding Student Scholarship

2021-2023

Korea University

 REFERENCES

Mohit Bansal Professor at UNC Chapel Hill  mbansal@cs.unc.edu

Jinkyu Kim Professor at Korea University  jinkyukim@korea.ac.kr

Sung Ju Hwang Professor at KAIST  sjhwang82@kaist.ac.kr

Yousung Park Emeritus professor at Korea University  yspark433@gmail.com